

KRA-DS-02: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)] I

Date: 2025-02-20

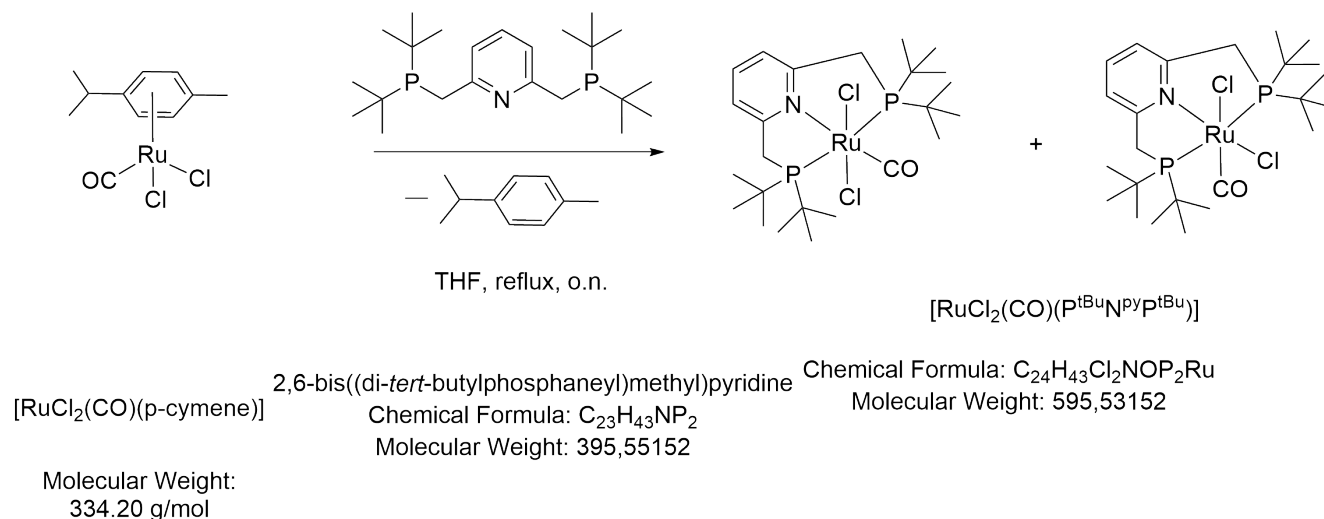
Tags: [RuCl₂(CO)(p-cymene)] Dichloro PNP [RuCl₂(CO)(PNP)] P(tBu)N(py)P(tBu) NMR 1H 31P KRA-DS

Category: Two-photon

Status: Done

Created by: David Schneider

Reaction scheme/sample structure



Literature/reference experiments

Literature	https://doi.org/10.1039/D1EE01053K
Reproduction	/
Related experiments	Experiment - AE-157: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)] Experiment - AE-230: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)] Experiment - AE-232: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)] Experiment - AE-261: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)]

Reagents

Name	Amount [mmol]	Equivalents	Purity	Mass _{theo,pure} [mg]	Mass _{theo} [mg]	Mass _{exp} [mg]	Molar mass [g/mol]	Volume [ml]
[RuCl ₂ (CO)(p-cymene)] (Two-photon - AE-444: Synthesis of [RuCl ₂ (CO)(p-cymene)])	1.19	1.00	0.80	398.37	497.96	497.86	334.20	/
PNP, 2,6-Bis(N,N-(tert-butyl)phosphinomethyl)pyridine (Experiment - AE-253: Synthesis of 2,6-Bis(P,P-di(tertbutyl)phosphinomethyl)pyridine)	1.28	1.25	0.90	507.6	564	564	395.55	/
THF (Prep work - AE-397: Degassing of THF)	/	/	/	/	/	/	/	50

Work-up and Analytical Reagents

Name	CAS Number / Experiment Number	Inventory number	Mass _{exp} [g]	Volume [ml]	Concentration [M]
n-heptane	AE-315: Degassing of n-Heptane	/	/	30	/
DCM-d2	Prep work - AE-410: Degassing of DCM-d2	/	/	0.6	/

Procedure/observations

All steps, unless mentioned otherwise, were carried out under argon using standard Schlenk technique or in an argon filled glovebox. (see Protocol [Protocol - Schlenk Technique](#))

Date	Time	Step	Observations
19.02.	15:40	A 100 mL and a 25 mL Schlenk flask was prepared	
	16:15	An air condenser was set under vacuum for approx. one hour	
	16:00	PNP(tBu) (564 mg) was weighted in the Glovebox K004 into the 100 mL Flask	white solid
	16:20	[RuCl ₂ (CO)(p-cymene)] (597.86 mg) was weighted in the 25 ml Flask	brick red powder
	17:30	To the flask which contained the [Ru] 15 mL of THF were added	dark red suspension; Ru.jpg
		To the flask containing the PNpTBu 15 mL of THF were added. The solution was stirred and shaken for 3 min.	suspension; PNpTBu.jpg
		The PNpTBu solution was added to the reaction flask in one go	
		The 25 mL flask was washed with 5 mL of THF because some solid (PNpTBu) stuck to the wall, The suspension was then also added to the reaction flask in one go	
		15 mL of THF was added to the reaction.	dark red solution
	18:05	The reaction mixture was stirred at rt and the flask was connected to the cooler in argon counterflow.	

	18:10-18:22	The reaction mixture was heated to reflux	reflux.jpg
	18:22 -	The reaction mixture was heated at reflux for approx. 13:40 h at 300 rpm.	
20.02.	- 10:02	The reaction mixture was cooled down to r.t.	after_reflux.jpg dark red suspension, precipitate
	10:50-11:35	The volume of the remaining reaction mixture was reduced under reduced pressure, using external cooling trap to approx. 25 mL	destillation.jpg reduced.jpg
	13:30	To the solution n-Heptane (18 mL) was added under stirring.	with_heptane.jpg more precipitation
		The obtained mixture was filtered according to Protocol - Filtration with frit technique and the residue was washed with n-Heptane (7 mL + 5 mL)	filtration.jpg
	15:00-16:55	The obtained solid was dried under reduced pressure. KRA-DS-02-1 Liquid phase: KRA-DS-02-2	dry_frit.jpg yellow-brown solid
25.02	15:00	The solid was transferred into vial in a 10 mL Flask	product.jpg
	15:45	A NMR-sample was prepared from KRA-DS-02-1 according to [Protocol] Preparation of NMR Sample in CD2Cl2.	

Analysis

Date	Time	Sample name	Analysis method	Used device	Solvent	Raw Data	Processed Data	Comparative data	Interpretation
25.02.	23:13 23:25	KRA-DS-02-1	¹ H, ³¹ P{ ¹ H}	IAAC 400 MHz I	DCM-d2	2025-02-26_13-01-44.zip	2025-02-26_13-01-44.zip_KRA-DS-02_10.nmrium	Experiment - AE-230: Synthesis of [RuCl ₂ (CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl ₂ (CO)(p-cymene)] -2	Desired product at 65 ppm (31P). All signals in ¹ H found. Purity according to ³¹ P and ¹ H of over 95 %
07.04.	/	KRA-DS-02-K	XRD	/	/	Kristallstruktur-KRA-DS-02-K.zip	/	/	60% cis, 15% trans, 25% Ru(III)

Product characterization

	mass [mg]	purity [%]	mass _{pure} [mg]	amount [mmol]	Yield [%]	
KRA-DS-02-1	407.63	>95 %	407.63	0.68	58	yellow-brown solid, product.jpg

Linked experiments

- JSC-KS-04: Degassing of n-Heptane
- AEI-047: Synthesis of 2,6-Bis(P,P-di(tertbutyl)phosphinomethyl)pyridine
- AEI-057: Synthesis of [RuCl₂(CO)(p-cymene)]
- AEI-060: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)]
- AEI-072: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)]
- AE-100: Synthesis of [RuCl₂(CO)(p-cymene)]
- AE-101: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)]
- AE-102: Degassing of THF
- AE-108: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)]
- AE-120: Degassing of n-Heptane
- AE-157: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)]
- KRA-080: Degassing of THF
- AE-221: Synthesis of [RuCl₂(CO)(p-cymene)]
- AE-230: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)]
- AE-232: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)]
- AE-248: Synthesis of [RuCl₂(CO)(p-cymene)]
- AE-253: Synthesis of 2,6-Bis(P,P-di(tertbutyl)phosphinomethyl)pyridine
- AE-261: Synthesis of [RuCl₂(CO)(PNP)] with 2,6-Bis((tert-butyl)phosphinomethyl)pyridine via [RuCl₂(CO)(p-cymene)]
- AE-315: Degassing of n-Heptane

Prep work - [AE-397: Degassing of THF](#)

Prep work - [AE-410: Degassing of DCM-d2](#)

Two-photon - [AE-444: Synthesis of \[RuCl2\(CO\)\(p-cymene\)\]](#)

Linked resources

Protocol - [Preparation of NMR Sample](#)

Protocol - [Filtration with frit technique](#)

Protocol - [Schlenk Technique](#)

Attached files

Kristallstruktur-KRA-DS-02-K.zip

sha256: 0d7f01d6b3a5251b8aee1ba4638e3b9bb7aca9ab53d5efc8c1764deb1d883fa4

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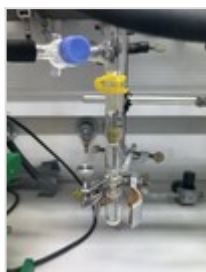
product.jpg

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dry_frit.jpg

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filtration.jpg

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with_heptane.jpg

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reduced.jpg

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distillation.jpg

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after_reflux.jpg

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reflux.jpg

sha256: c949a680b61df2c28c92b352f2127a5f27944e81cd2d26c84e6727ca719ce593



PNPtBu.jpg

sha256: f09bb6dc336f25bcbc092d4f633794228f3e7ba1fb51211dad5d36cf8f080a95



Ru.jpg

sha256: 4c1c3f29051ae738359754460f44a2972e8b68d24132a8cab92c716210966358



Comment

On 2025-02-24 09:39:47 Kristína Rabatinová wrote:

Notes:

- * adding the analytical reagents table
- * 19.02., 15:40, "A 100 ml and a 25 ml Schlenk flask were prepared." should be grey
- * 19.02. How was the PNPtBu solution added to the reaction flask? --> adding dropwise or at once, adding under stirring ALSO here changing solution to suspension
- * "The 25 ml flask was washed with 5 ml of THF," --> I would add that this was done because some solid (PNPtBu) stuck to the flask
- * "15 ml of THF was ..." --> was it a dark red solution or a suspension?
- * 19.02. 18:22 - "the reaction mixture was heated AT reflux ..." since heated TO reflux is above
- * again generally add more descriptions to the picture, are a "rule" every picture should have a description, How does the solution changes after reflux? Darker/brighter or not really? Is it a solution or a suspension? Do we see a precipitate? What happens when we add heptane? More precipitation? Color of the precipitation and the solution? How does the solid looks like on the frit? Color?



Unique eLabID: 20250220-68b2515edd676728dafdc65ef3c068d01953260f
Link: <https://elab.water-splitting.org/experiments.php?mode=view&id=1776>